Care Mate

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Abstract

This project presents a healthcare solution that includes two systems: Care Signal, a device for emergency alerts, and Care Mate, a program for daily patient care management. Care Signal helps users send help requests quickly and easily, while Care Mate allows caregivers to track the patient's health, schedule medications, and manage appointments. Although the two systems are currently separate, their future integration aims to improve the quality of care using simple and effective technology.

Introduction

As the need for smart and fast healthcare increases, the Care Signal device was developed to offer a simple way to request help with one button.

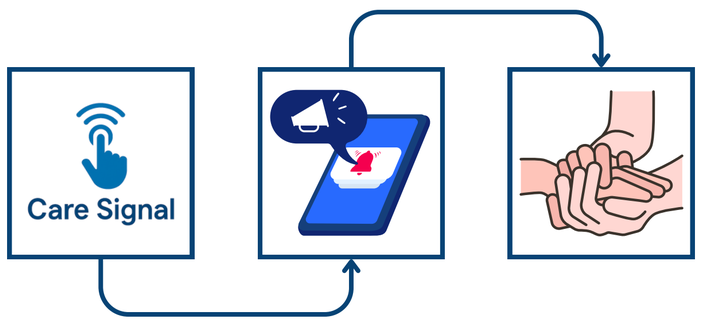
To support daily care and documentation, the Care Mate system was created to help caregivers manage patient records, add medical notes, and organize medications and appointments.

This project provides a modern example of healthcare using simple and user-friendly technologies.

Methodology

The project was developed in two main phases:

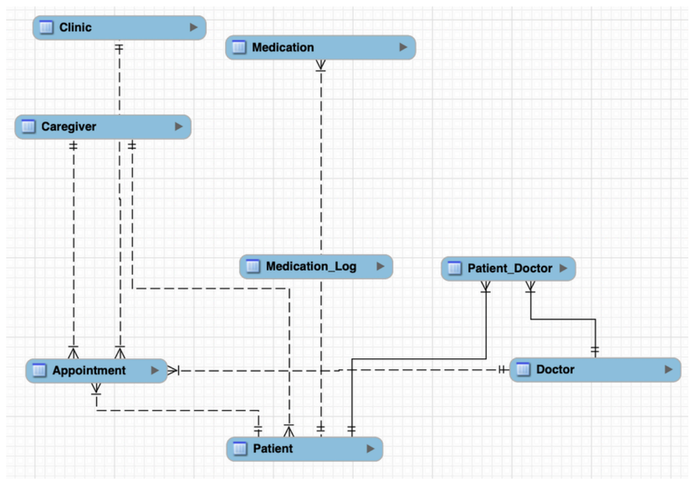
Phase 1 – Care Signal:
A simple emergency alert device was designed using Arduino and a GSM module. When the patient presses the button, the device sends a direct alert to the caregiver, who can respond immediately. This approach helps ensure quick and easy communication during emergencies. As shown in Fig [1].



[Fig 1] Care Signal process from alert to caregiver response

Methodology

Phase 2 – Care Mate:
As a complement to the device, an electronic care system was developed to help caregivers manage daily tasks such as recording notes, scheduling appointments, and tracking medications. Built with Java and MySQL, it also allows doctors to view patient records and add notes when needed. Fig [2] shows the system's database structure and relationships.



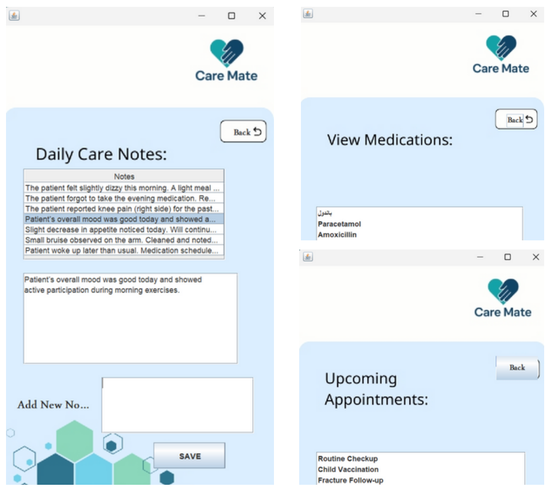
[Fig 2] Care Mate Database framework

Results and Discussion

The Care Signal device proved effective in providing a fast and simple way to send emergency alerts, especially for elderly individuals and people with special needs. The alert is sent directly to the caregiver using GSM technology, ensuring a quick response in critical situations. It showed strong performance in real-life simulation scenarios.

Results and Discussion

At the same time, the Care Mate system offered useful tools for managing daily care. It allows caregivers to record medical notes, track medications, and organize appointments as shown in Fig [3]. The system keeps patient data structured and accessible, helping maintain consistent care, especially for chronic or complex cases.



[Fig 3] Care Mate User Interface

Conclusion

This project shows how simple technologies like Care Signal and Care Mate can improve healthcare. The device provides emergency alerts, and the system helps manage daily care. The planned integration aims to create a smarter and more connected home care experience.

References